EXPERIMENT 3

VEHICLE SERVICE REQUEST MANAGEMENT SYSTEM

A Survey Report on Digital Service Booking, Request Tracking, and Workshop Workflow Automation

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ACADEMIC SURVEY REPORT

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Note: This report presents a proposed academic system design. Features and outcomes described are intended design goals, not claims of a deployed production service.

1. Introduction

Vehicle maintenance centres handle a steady flow of customer enquiries, service bookings, repair requests, inspections, and follow-up communication. When these activities are recorded through phone calls, paper forms, or disconnected spreadsheets, information can become difficult to track. Customers may not know whether a request has been accepted, while service staff may have to spend time searching for vehicle details or confirming appointment status.

The Vehicle Service Request Management System is a proposed digital application that organizes service requests from initial submission through appointment scheduling, work assignment, status updates, and closure. The system provides a common record for customers and workshop staff, helping make the process clearer, more consistent, and easier to monitor.

The purpose of this survey is to examine the need for a structured vehicle-service workflow, identify the system's key modules, and outline practical considerations for implementing a secure and usable application.

1.1 Objectives

To provide a single point for registering vehicle service requests and customer details.

To support appointment scheduling, request prioritization, and service-status tracking.

To improve coordination between customers, service advisors, technicians, and administrators.

To maintain searchable service histories and produce basic operational reports.

1.2 Methodology

This report uses a conceptual review of common vehicle workshop activities and software application design principles. The workflow, data fields, and modules are proposed for an academic prototype; they should be validated with actual workshop users before deployment.

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2. Understanding the System

A Vehicle Service Request Management System captures a customer's maintenance or repair requirement and turns it into a trackable service record. A request may concern routine servicing, an engine or brake issue, tyre replacement, battery inspection, a warning light, or another concern reported by the vehicle owner.

The application is not intended to diagnose a vehicle automatically. Instead, it helps record the customer's description, vehicle information, preferred appointment, and subsequent workshop updates. A qualified service advisor or technician remains responsible for inspecting the vehicle, confirming the work required, and documenting the outcome.

2.1 Typical Users

Customer: submits a request, selects a preferred time, and checks progress.

Service advisor: reviews requests, contacts customers, confirms appointments, and records estimates.

Technician: views assigned jobs and updates work progress or inspection notes.

Administrator: manages users, service categories, schedules, and operational reports.

2.2 Core Components

Request form and customer/vehicle profile; appointment calendar; request queue and assignment view; service-status timeline; notification and communication log; service-history database; reporting dashboard; role-based access control.

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3. Current Process and Need for Digitization

In a manual process, customers may call or visit the workshop to describe a problem. Staff write the details in a register or spreadsheet, then separately coordinate appointment slots and technician availability. If the customer calls again, an employee may need to locate the original entry or ask for the same information again. Updates can be delayed when several staff members maintain separate records.

A digital request system can reduce this fragmentation by giving each service request a unique reference number and a clearly defined status. It can also make the work queue visible to authorized staff, helping them identify pending requests, upcoming appointments, and jobs that need customer confirmation.

3.1 Common Process Challenges

Repeated data entry and incomplete vehicle or contact details.

Unclear ownership of a request after it has been received.

Appointment conflicts caused by limited visibility of workshop capacity.

Customers having to contact the workshop repeatedly for progress updates.

Difficulty reviewing earlier repairs or preparing service summaries.

3.2 Requirements for an Improved Process

Use a standard request form with required fields and validation.

Record each status change with a timestamp and responsible user.

Separate request submission from appointment confirmation.

Allow staff to assign jobs according to skills, availability, and workload.

Keep service records searchable while limiting access to authorized users.

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4. Proposed System and Core Features

The proposed application follows a request-centred design. Each request links a customer, a vehicle, a service category, a preferred time, and a status history. The interface should remain simple for customers while offering staff a more detailed operational view. The following features form a practical baseline for an academic implementation.

4.1 Customer and Vehicle Registration

Capture customer name, contact details, vehicle registration number, make/model, and relevant vehicle notes. Reusable profiles reduce the need to re-enter the same information for future visits.

4.2 Service Request Submission

Let customers choose a service category, describe the issue, indicate urgency, and provide a preferred date or time. The system validates mandatory fields and creates a unique request ID after submission.

4.3 Appointment Scheduling

Display available slots and allow staff to confirm, reschedule, or cancel appointments. A request should not be treated as a confirmed booking until the workshop has accepted the slot.

4.4 Status Tracking and Notifications

Use clear stages such as Submitted, Under Review, Appointment Confirmed, Vehicle Received, In Service, Awaiting Approval, Completed, and Cancelled. Updates can be shown in the application and, where configured, sent through email or SMS.

4.5 Staff Assignment and Service History

Authorized staff can assign a request to a service advisor or technician, record inspection notes, and maintain a history of work performed. Customers can refer to previous service records where permissions allow.

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5. Workflow and Functional Modules

A consistent workflow helps ensure that requests are not lost between submission and completion. The system should support both the normal path and exceptions, such as a customer changing an appointment or a workshop discovering that additional approval is required.

5.1 Request Lifecycle

1. Customer enters contact and vehicle details, describes the service need, and submits the request.

2. The system validates the form and generates a unique request reference.

3. A service advisor reviews the request, checks capacity, and contacts the customer if clarification is needed.

4. The appointment is confirmed or an alternative slot is proposed.

5. On vehicle arrival, staff update the request and assign the job for inspection or service.

6. Staff record findings, estimate or approval needs, work progress, and completion details.

7. The request is closed after the work and handover are recorded; the service history remains available for future reference.

5.2 Functional Modules

Customer module: profile, vehicle details, request submission, booking view, and status timeline.

Service desk module: request review, appointment management, customer communication, and job assignment.

Technician module: assigned jobs, inspection checklist, work notes, parts or labour notes, and completion update.

Administration module: user roles, service categories, working hours, slot capacity, and reports.

Reporting module: request volume, service categories, pending jobs, cancellations, and completion time.

5.3 Example Request Record

Request ID: VSR-000124; Vehicle: TN-XX-1234; Service category: General servicing; Preferred slot: 14 October, 10:00 AM; Status: Appointment Confirmed; Assigned advisor: Service Desk. These values are illustrative and should be replaced with actual prototype data.

6. Data Management, Security, and Validation

The quality of the system depends on accurate records and appropriate access controls. A relational database is suitable for linking customers, vehicles, requests, appointments, staff assignments, and status updates while maintaining consistency between related records.

6.1 Suggested Data Entities

Customer: customer ID, name, phone/email, and communication preference.

Vehicle: vehicle ID, customer ID, registration number, make/model, and optional year.

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Service Request: request ID, vehicle ID, category, issue description, urgency, creation time, and current status.

Appointment: appointment ID, request ID, date/time, booking status, and assigned service bay where applicable.

Staff and Assignment: staff ID, role, skill category, and request assignment.

Status History: request ID, previous status, new status, timestamp, updated by, and optional note.

6.2 Validation and Data Quality

Require essential customer, vehicle, and request information before submission.

Validate contact formats and prevent duplicate or malformed identifiers where appropriate.

Check appointment slots before confirming a booking.

Use controlled status values instead of free-text status entries.

Keep timestamps and a change history for important actions.

6.3 Security and Privacy

Apply role-based permissions so customers can see only their own requests and staff can access information needed for their duties.

Protect credentials, use secure connections in deployment, and avoid storing unnecessary personal information.

Provide backup and recovery procedures, session management, and an audit trail for sensitive updates.

Define a retention policy for customer and vehicle records and ensure that users understand how their information is used.

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7. Technology and Implementation Landscape

The application can be implemented using a low-code platform or a conventional web stack. The final choice should depend on the academic requirements, available skills, integration needs, and deployment constraints. The architecture should keep the user interface, application logic, and data storage logically separated.

7.1 Possible Implementation Options

Low-code approach: a platform such as Zoho Creator can provide forms, reports, workflows, permissions, and mobile-friendly access with less custom code. Deluge scripts may be used for validations and business rules.

Custom web approach: a frontend built with HTML, CSS, JavaScript, or React can communicate with a backend such as Node.js or Flask. A relational database such as MySQL or PostgreSQL can store requests and service history.

Both approaches can support a dashboard for staff, request forms for customers, and reporting. The system should not depend on a particular technology unless required by the course.

7.2 High-Level Architecture

Presentation layer: customer request page, booking interface, staff dashboard, and administration screens.

Application layer: request validation, appointment rules, status transitions, notifications, and permission checks.

Data layer: customer, vehicle, request, appointment, staff, and status-history records.

Integration layer (optional): email/SMS services, payment services, or workshop inventory systems, subject to access and testing.

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7.3 Testing Plan

Functional testing: verify request creation, appointment changes, status updates, and report generation.

Validation testing: submit missing fields, invalid contact details, and conflicting appointment times.

Access testing: confirm that each role sees only permitted functions and records.

Usability testing: ask representative users to complete common tasks and note confusion or unnecessary steps.

Recovery testing: verify that backups can be restored and that failed notifications do not erase the service request.

8. Benefits, Limitations, and Risks

A structured service request application can improve visibility and consistency, but its effectiveness depends on staff adoption, accurate data entry, and a workflow that reflects real workshop practices.

8.1 Expected Benefits

Centralized customer, vehicle, and service-request records.

Clearer ownership of requests and fewer missed follow-ups.

Improved visibility into appointment availability and pending work.

Faster retrieval of service history and easier preparation of reports.

More consistent communication through recorded status updates.

8.2 Limitations and Risks

Incorrect information entered by users can still lead to scheduling or service errors.

Staff may continue using parallel paper records unless training and process changes are provided.

External messaging or payment integrations may add cost and configuration effort.

Vehicle diagnosis, repair estimates, and safety decisions still require qualified human review.

System availability, data protection, and backup arrangements must be addressed before real-world use.

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9. Future Scope and Recommendations

The initial version should prioritize reliable request capture, appointment handling, status tracking, and role-based access. Advanced features should be introduced only after the core workflow is tested with users.

9.1 Future Enhancements

Automated reminders for upcoming appointments and requests awaiting customer approval.

Service reminders based on date, mileage, or recorded maintenance history.

Customer feedback and service-quality surveys after job completion.

Parts availability integration to help staff plan jobs more accurately.

Analytics for peak booking periods, common service categories, and average turnaround time.

Mobile-friendly staff views for updating work progress from the service area.

9.2 Recommendations

Map the existing workshop workflow before designing forms and statuses.

Use clear labels and avoid collecting information that is not needed for the service.

Pilot the application with a small group of staff and customers before wider rollout.

Define ownership for request updates, appointment changes, and customer communication.

Review access permissions, backups, and data-retention rules as part of implementation rather than after deployment.

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10. Conclusion

The Vehicle Service Request Management System offers a structured way to manage service enquiries, vehicle details, appointments, workshop assignments, and progress updates. By maintaining a unique record for each request and recording its lifecycle, the proposed system can make the service process easier to follow for customers and staff. Its value depends on practical workflow design, reliable data, appropriate security, and consistent use by the workshop team.

For an academic prototype, the recommended starting point is a small but complete workflow: submit a request, review it, confirm an appointment, update its status, and close the service record. This provides a foundation that can later be extended with reminders, reporting, and integrations.

11. About the Author

This report was prepared by Suhritha Sri, a student of Computer Science and Business Systems (CSBS), Roll No. 241401110. The report presents a proposed digital workflow for organizing vehicle service requests and related workshop activities.

12. References

- Oracle Academy, Application Development Foundations learning materials.
- Oracle APEX Documentation, application development and SQL Workshop concepts.

• Zoho Creator Documentation, forms, workflows, reports, and application permissions.
• OWASP Foundation, guidance on access control, secure application design, and data protection.
• General software engineering practices for requirements analysis, database design, functional testing, and user acceptance testing.

End of Report

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